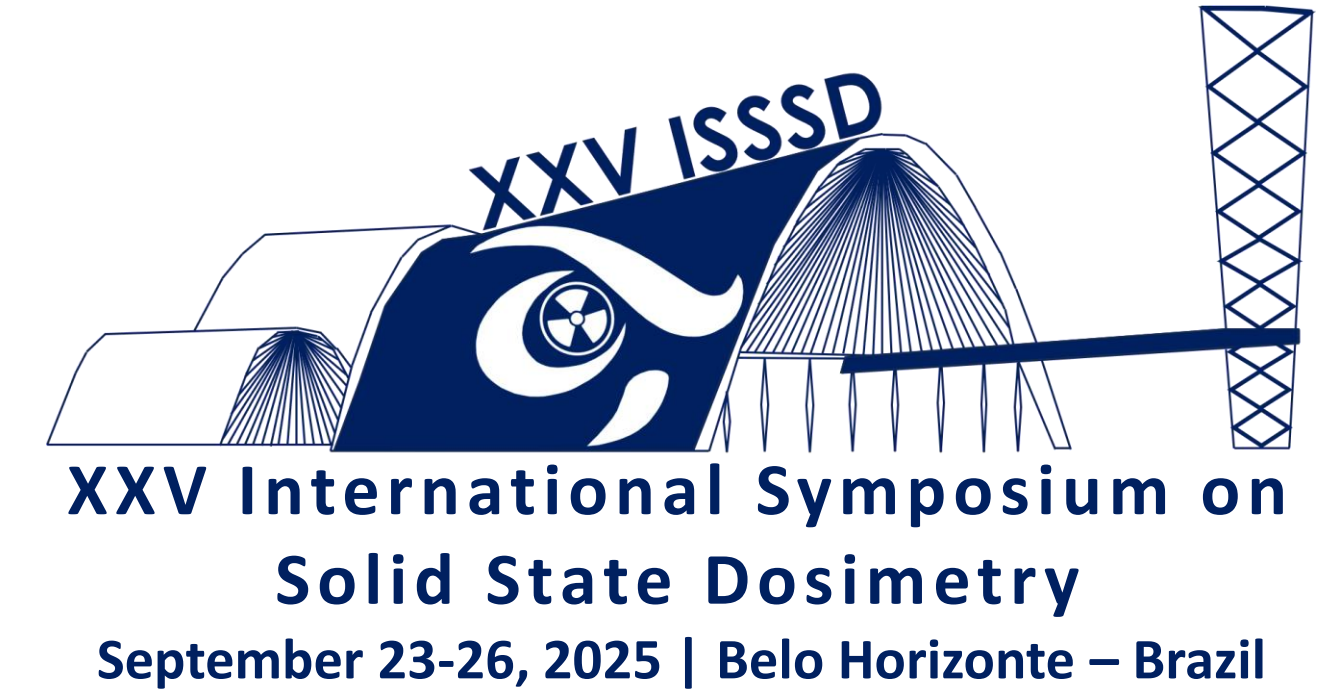


Development of a Dedicated System for Generating Input Files for the MCNP and Adaptation of a Phantom with Edema for Simulation of Prostate Brachytherapy with ^{125}I

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1. INTRODUCTION

Prostate cancer is the leading cause of death from neoplasms among men in Brazil. Low-dose-rate (LDR) brachytherapy using radioactive iodine-125 (^{125}I) seeds can cause prostatic edema, which leads to seed displacement and changes in dose distribution. To overcome the difficulty of quantifying this variation, a computer system was developed that can generate input files for the MCNP code, based on the original treatment plan and the predicted edema. This approach allows the impact of prostatic swelling on the actual dose administered to be evaluated.

2. MATERIALS AND METHODS

2.1 The specific features of brachytherapy

Low dose rate (LDR) brachytherapy is a minimally invasive cancer treatment in which radioactive ^{125}I seeds are permanently inserted into the tumor site. The procedure, guided by ultrasound, is performed by a multidisciplinary team, with emphasis on the role of the physician and medical physicist in planning.

2.2 ^{125}I seeds

The Amersham model 6711 ^{125}I seed, used as a reference in this study, was modeled as follows: an internal silver cylinder containing a $1\mu\text{m}$ -thick ^{125}I emitting film. This assembly is externally encapsulated by a titanium cylinder, 0.81 mm in diameter and 4.5 mm in length, and is closed by two ellipsoids.

2.3 Java and NetBeans IDE

The software was developed (Figure 1) in Java using NetBeans IDE, which ensured a robust and modular platform. Its structure, consisting of 24 files, optimizes code maintenance. To function, the system uses different modules:

Calculation modules: perform calculations of physical and geometric parameters.

MCNP modules: generate input files for the MCNP code (Figure 2).

Reporting and storage modules: record usage data and specifications in an SQL database.

Phantom modification module: adapts the 3D model to the scenario.

Results extraction module: extracts data for post-simulation analysis.

Registration modules: manage user access.

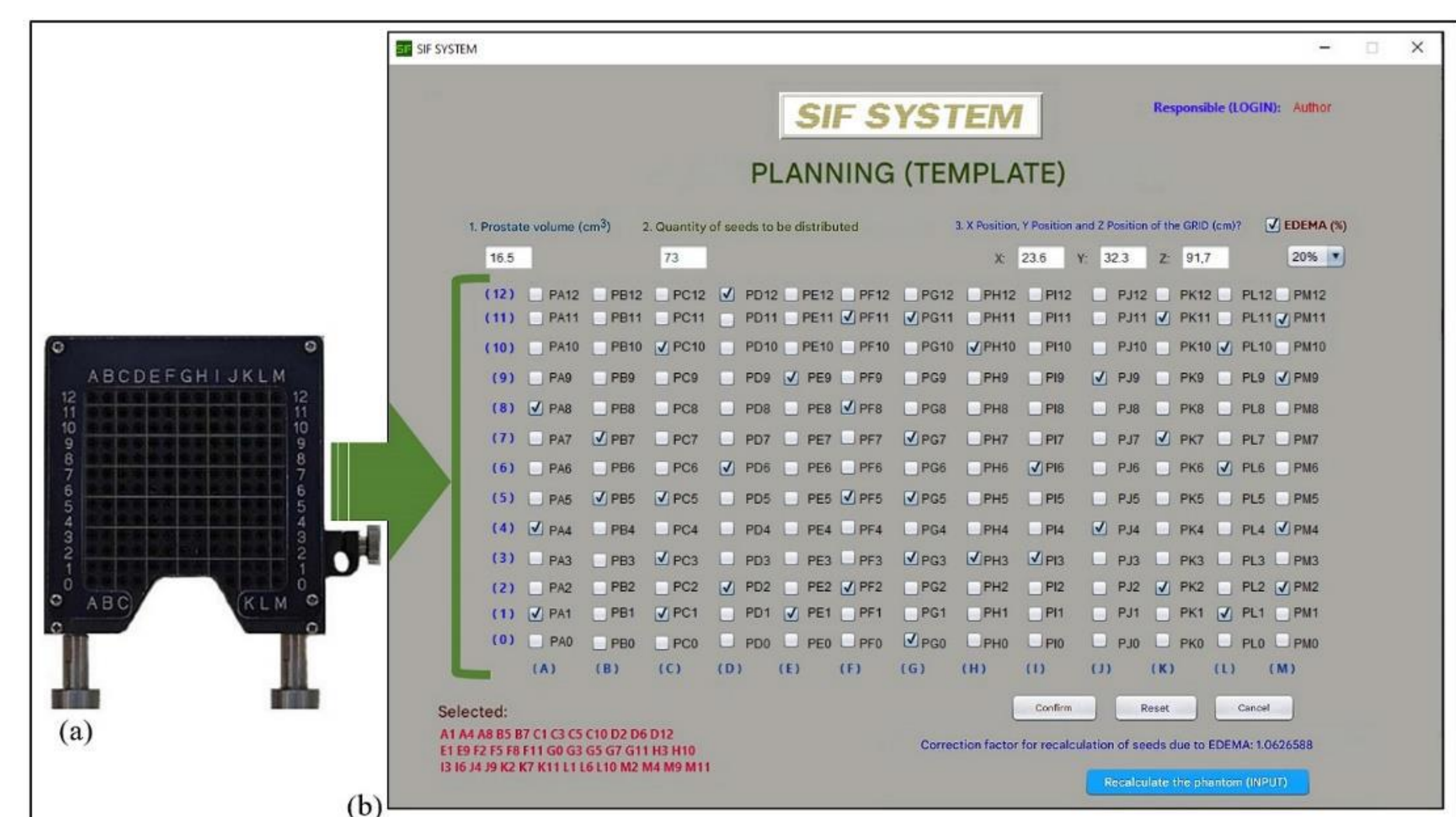


Figure 2: (a) The template used in the clinical procedure. (b) The computer system and the correlation with the template.

3. RESULTS

To validate the system, a voxel phantom with a 16.5 cm^3 prostate and 73 ^{125}I seeds was used to deliver 144 Gy. In the 50% edema scenario, the system adjusted seed positions, and the MCNPX simulation showed a dose drop to $\sim 125.28\text{ Gy}$ (87%). Figure 3 (DVH) illustrates the effect of edema on dose distribution.

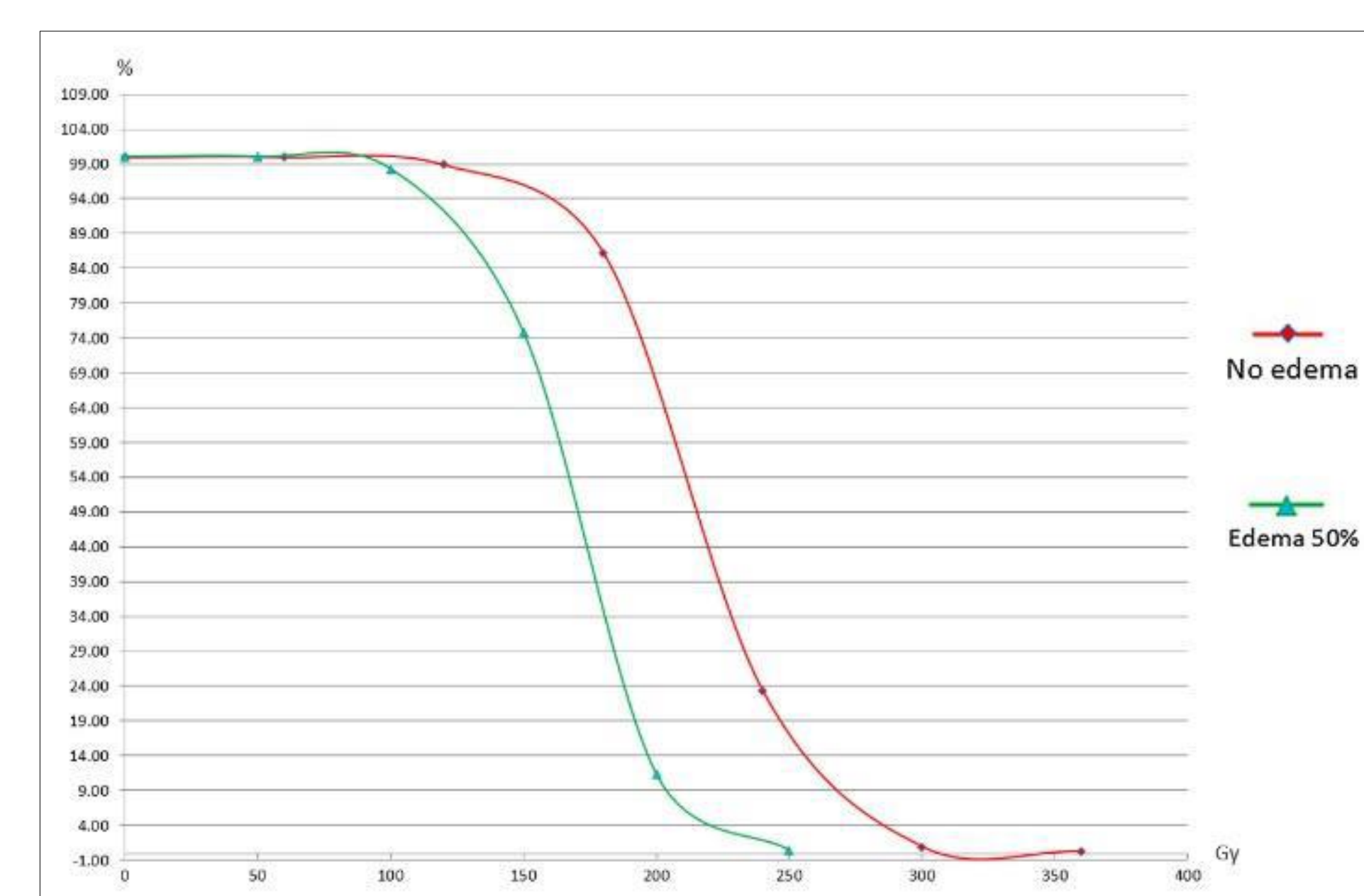


Figure 3: Dose-volume histogram (DVH) comparing dose distribution with and without edema.

The ptrack file generated by the simulation enabled 2D and 3D visualization of photon trajectories and secondary events in organs like the rectum and bladder, using Moritz (Figure 4).

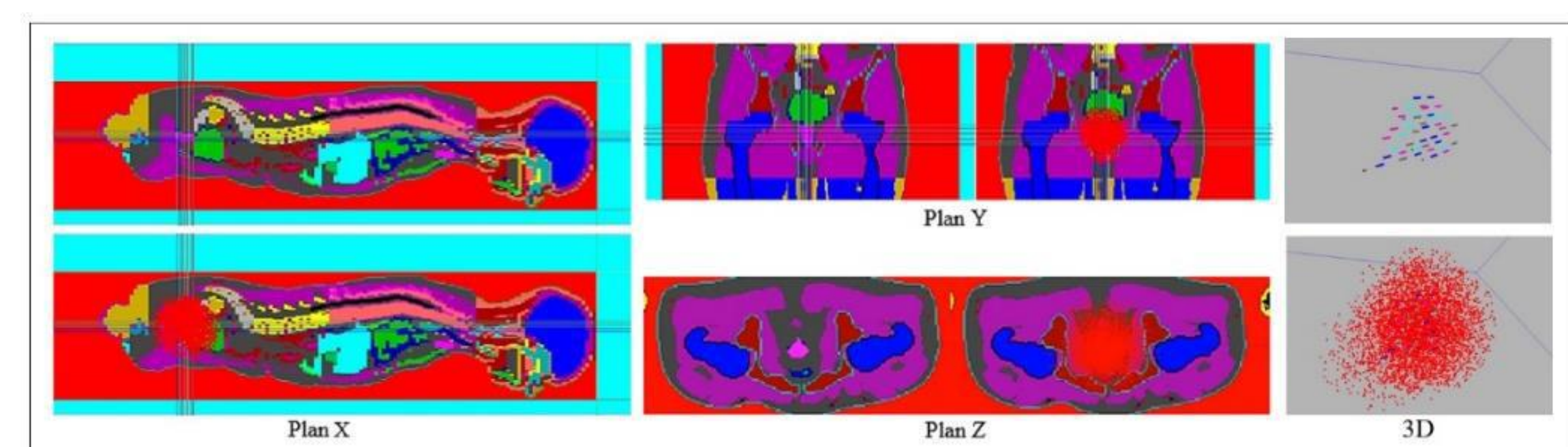


Figure 4: Animation capture of the X, Y, Z, and 3D planes in Moritz Geometry Tool, based on the ptrack file from the MCNPX simulation.

4. CONCLUSIONS

This work advances the simulation of LDR brachytherapy by developing a system that automates the generation of MCNP files considering prostatic edema, bringing simulations closer to clinical reality. The system, developed in Java, reduces model preparation time, allowing greater focus on physical analysis and paving the way for future integration with artificial intelligence techniques, provided that a robust database containing brachytherapy plans is built.



Figure 1: Overview of the system developed to generate input files.